FloPoCo Installation and IP Generation

This project uses **FloPoCo** to generate floating-point hardware IP blocks.

The generated VHDL files are **not stored in this repository**. Instead, each user generates the required IP locally.

Requirements

- Docker
- Git

Ubuntu

Install Docker:

```
sudo apt update
sudo apt install docker.io
sudo systemctl enable docker
sudo systemctl start docker
sudo usermod -aG docker $USER
```

Log out and back in (or reboot) so the Docker group membership takes effect.

Verify the installation:

```
docker run hello-world
```

Install FloPoCo

Clone the FloPoCo repository:

```
git clone https://gitlab.com/flopoco/flopoco.git
cd flopoco
```

Build the Docker image:

```
make docker DOCKER_IMAGE=debian DOCKER_BRANCH=master
```

Create a shell function

Add the following function to your `~/.bashrc` or `~/.zshrc`:

```
flopoco() {
    docker run --rm
        -v "$PWD":/flopoco_workspace
        flopoco:debian-5.0.0 "$@"
}
```

Reload your shell:

```
source ~/.bashrc
```

or

```
source ~/.zshrc
```

Generate IP

Navigate to the directory where you want the generated VHDL files to be placed.

Example: IEEE754 single-precision floating-point adder

```
mkdir fp_ip
cd fp_ip

flopoco FPAdd we=8 wf=23
```

The generated VHDL files will appear in the current working directory.

Notes

- `we=8 wf=23` corresponds to IEEE754 single precision.
- Generated VHDL files are intentionally excluded from version control.
- If different floating-point operators or precisions are required, regenerate them using FloPoCo with the desired parameters.

For more information, refer to the official FloPoCo documentation.